If the object is at infinity, the image is formed at the focus of the lens. Thus, for $u = \infty$, $v = f$. Then the equation becomes.

$$\frac{1}{f} - \frac{1}{\infty} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad (6.61)$$

If the lens is kept in air, then we can take $n_1 = 1$ and $n_2 = n$. So the equation (6.61) becomes,

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad (6.62)$$

The above formula is called as the ***lens maker's formula***, because it tells the lens manufactures what curvature is needed for a material of particular refractive index to make a lens of desired focal length. This formula holds good also for any type of lens. By comparing the equations (6.60) and (6.61) we can write,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \qquad (6.63)$$

The above equation is known as ***lens equation*** which relates the object distance u and image distance v with the focal length f of the lens. This equation holds good for any type of lens.

6.6.4 Lateral magnification in thin lens

Let us consider an object OO' of height h_1 placed on the principal axis with its height perpendicular to the principal axis. The inverted real image II' is formed which has a height h_2 as shown in Figure 6.35.

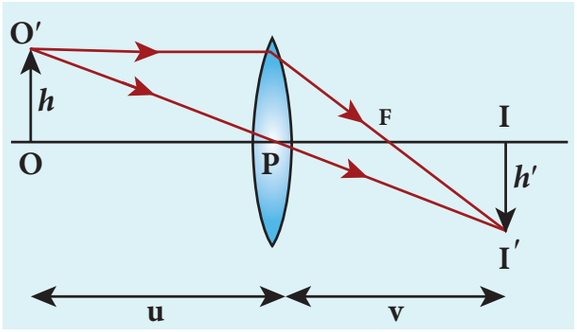


Figure 6.35 Lateral magnification in thin lens

The lateral (or) transverse magnification m is defined as the ratio of the height of the image to height of the object.

$$m = \frac{II'}{OO'} \qquad (6.64)$$

From the two similar triangles $\Delta POO'$ and $\Delta PII'$, we can write,

$$\frac{II'}{OO'} = \frac{PI}{PO} \qquad (6.65)$$

On applying sign convention,

$$\frac{-h'}{h} = \frac{v}{-u}$$

Substituting this in the equation (6.65) for magnification,

$$m = \frac{-h'}{h} = \frac{v}{-u}$$

After rearranging,

$$m = \frac{h'}{h} = \frac{v}{u} \qquad (6.66)$$

The magnification is negative for real image and positive for virtual image. In the case of a concave lens, the magnification is always positive and less than one.

We can also have the other forms of equations for magnification by combining the lens equation as,

$$m = \frac{h'}{h} = \frac{f}{f+u} \text{ (or) } m = \frac{h'}{h} = \frac{f-v}{f} \qquad (6.67)$$

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